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SEMICONDUCTOR DEVICES AND METHODS OF MANUFACTURING THEREOF

Abstract

A semiconductor device includes a plurality of first channel layers vertically spaced from one another and a plurality of second channel layers vertically spaced from one another. Each of the plurality of first and second channel layers extend along a first lateral direction. The semiconductor device includes an isolation structure disposed between the plurality of first channel layers and the plurality of second channel layers along a second lateral direction perpendicular to the first lateral direction. The semiconductor device includes a plurality of inner spacers discretely disposed along a first sidewall of the isolation structure that faces toward the first lateral direction, or discretely disposed along a second sidewall of the isolation structure that faces away from the first lateral direction wherein an interface between each of the plurality of inner spacers and the first or second sidewall has a vertical profile.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of U.S. Utility application Ser. No. 17/672,404, filed on Feb. 15, 2022, which is a continuation-in-part of U.S. Utility application Ser. No. 17/460,204, filed on Aug. 28, 2021, the entire contents of each of which are incorporated herein by reference for all purposes.

BACKGROUND

[0002] The semiconductor industry has experienced rapid growth due to continuous improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, this improvement in integration density has come from repeated reductions in minimum feature size, which allows more components to be integrated into a given area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates a perspective view of a gate-all-around (GAA) field-effect-transistor (FET) device, in accordance with some embodiments.

[0005] FIG. 2 illustrates a flow chart of an example method for making a non-planar transistor device, in accordance with some embodiments.

[0006] FIGS. 3, 4A, 4B, 5A, 5B, 6, 7A, 7B, 8, 9A, 9B, 10A, 10B, 10C, 11A, 11B, 11C, 11D, 11E, 11F, 11G, 11H, 11I, 11J, 11K, 11L, 12A, 12B, 12C, 12D, 12E, 12F, 12G, 12H, 12I, 12J, 13A, 13B, 13C, 13D, 13E, 14A, 14B, and 14C illustrate cross-sectional or top views of an example GAA FET device (or a portion of the example GAA FET device) during various fabrication stages, made by the method of FIG. 2, in accordance with some embodiments.

[0007] FIGS. 15 and 16 illustrates a top view and a cross-sectional view of another portion of the example GAA FET, respectively, made by the method of FIG. 2, in accordance with some embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components

and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] In contemporary semiconductor device fabrication processes, a large number of semiconductor devices, such as silicon channel n-type field effect transistors (nFETs) and silicon germanium channel p-type field effect transistors (pFETs), are fabricated on a single wafer. Non-planar transistor device architectures, such as fin-based transistors, can provide increased device density and increased performance over planar transistors. Some advanced non-planar transistor device architectures, such as nanostructure transistors, can further increase the performance over fin-based transistors.

[0011] Example nanostructure transistors include nanosheet transistors, nanowire transistors, gate-all-around (GAA) transistors, and the like. The nanostructure transistor includes one or more nanostructures, collectively configured as a conduction channel of the transistor, that are fully wrapped by a gate structure. When compared to the fin-based transistors where the channel is partially wrapped by a gate structure, the nanostructure transistor, in general, can include one or more gate stacks that wrap around the full perimeter of a nanostructure channel. In some embodiments, the gate stack partially wrap around the nanostructure channel. As such, control over the nanostructure channel may be further improved, thus causing, for example, a relatively large driving current given the similar size of the fin-based transistor and nanostructure transistor.

[0012] An integrated circuit typically includes a large number of such nanostructure transistors disposed over a substrate. To configure these nanostructure transistors operating as certain circuits, some of the nanostructure transistors can be operatively connected to or disconnected from each other. For example, plural (intermediate) channel structures of nanostructure transistors may be formed over a substrate, which are arranged in parallel with one another. A (e.g., dummy) gate structure is next formed over these channel structures. Prior to replacing the dummy gate structure with an active (e.g., metal) gate structure, at least a portion of a middle one of the channel structures, which is designed to decouple side ones of the channel structures, is replaced with a dielectric material, thereby forming an isolation structure. As such, transistors, respectively formed based on the side channel structures, can be electrically decoupled from each other with the isolation structure.

[0013] In existing technologies, when removing the middle channel structure (or more specifically, the semiconductor layers of the middle channel structure), a number of inner spacers vertically disposed between adjacent semiconductor channel layers of the middle channel structure typically remain substantially intact. Such intact inner spacers can present a sidewall protruding toward a cavity formed by removing the middle channel structure. The cavity is then filled with a dielectric material to form an isolation structure. With these protruded inner spacers, a (e.g., lower) portion of the middle channel structure may not be removed as desired. For example, one or more of the

semiconductor layers of the middle channel structure relatively close to the bottom may remain. Stated another way, the cavity may have some portions of the semiconductor layers left. As such, the isolation structure, formed by filling the cavity with a dielectric material, may not successfully isolate the transistor formed by the side channel structures, respectively, as originally planned. [0014] Embodiments of the present disclosure are discussed in the context of forming non-planar transistor devices (e.g., nanostructure transistor devices), and in particular, in the context of forming an isolation structure between two adjacent channel structures. Different from the existing technologies, the disclosed method provides shaping the sidewalls of inner spacers on a side facing a (e.g., middle) channel structure between the two adjacent channel structures, in which the middle channel structure defines a footprint of the to-be-formed isolation structure. The step of shaping the sidewalls may be performed concurrently with removing the middle channel structure. In various embodiments, the sidewalls may be each shaped to present a nearly vertical profile. With a majority portion of the protruded sidewall being removed, it can be more efficiently to form the isolation structure without any semiconductor layers remaining therein. As such, the transistors, formed by the adjacent channel structures, respectively, can be more reliably decoupled from each other by the isolation structure.

[0015] FIG. 1 illustrates a perspective view of an example non-planar (e.g., GAA) transistor device **100**, in accordance with various embodiments. The non-planar transistor device **100** includes a substrate **102** and a number of semiconductor layers (e.g., nanosheets, nanowires, or otherwise nanostructures) **104** above the substrate **102**. The semiconductor layers **104** are vertically separated from one another, which can collectively function as a (conduction) channel of the GAA FET device **100**. Isolation regions **106** are formed on opposing sides of a protruded portion of the substrate **102**, with the semiconductor layers **104** disposed above the protruded portion. A gate structure **108** wraps around each of the semiconductor layers **104** (e.g., a full perimeter of each of the semiconductor layers **104**). A gate spacer **109** extends along each sidewall of the gate structure **108**. Source/drain structures are disposed on opposing sides of the gate structure **108** with the gate spacer **109** disposed therebetween, e.g., source/drain structure **110** shown in FIG. 1. An interlayer dielectric (ILD) **112** is disposed over the source/drain structure **110**.

[0016] The GAA FET device shown in FIG. 1 is simplified, and thus, it should be understood that one or more features of a completed GAA FET device may not be shown in FIG. 1. For example, the other source/drain structure opposite the gate structure **108** from the source/drain structure **110** and the ILD disposed over such a source/drain structure are not shown in FIG. 1. Further, FIG. 1 is provided as a reference to illustrate a number of cross-sections in subsequent figures. As indicated, cross-section A-A is cut along a longitudinal axis (e.g., a lengthwise direction) of the semiconductor (channel) layers **104** and in a direction of a current flow between the source/drain structures; and cross-section B-B is cut along a longitudinal axis (e.g., a lengthwise direction) of the gate structure **108**. In some arrangements, cross-section A-A and cross-section B-B may be perpendicular to each other. Subsequent figures refer to these reference cross-sections for clarity.

[0017] FIG. 2 illustrates a flowchart of a method **200** to form a non-planar transistor device, according to one or more embodiments of the present disclosure. For example, at least some of the operations (or steps) of the method **200** can be used to form a FinFET device, a GAA FET device (e.g., GAA FET device **100**), a nanosheet transistor device, a nanowire transistor device, a vertical transistor device, a gate-all-around (GAA) transistor device, or the like. It is noted that the method **200** is merely an example, and is not intended to limit the present disclosure. Accordingly, it is understood that additional operations may be provided before, during, and after the method **200** of FIG. 2, and that some other operations may only be briefly described herein. In some embodiments, operations of the method **200** may be associated with cross-sectional or top views of at least a portion of an example GAA FET device at various fabrication stages as shown in FIGS. 3, 4A, 4B, 5A, 5B, 6, 7A, 7B, 8, 9A, 9B, 10A, 10B, 10C, 11A, 11B, 11C, 11D, 11E, 11F, 11G, 11H, 11I, 11J, 11K, 11L, 12A, 12B, 12C, 12D, 12E, 12F, 12G, 12H, 12I, 12J, 13A, 13B, 13C, 13D, 13E, 14A,

14B, and **14C**, respectively, which will be discussed in further detail below.

[0018] In brief overview, the method **200** starts with operation **202** of providing a substrate. The method **200** continues to operation **204** of forming a first fin structure, a second fin structure, and a third fin structure. Each of the first to third fin structures includes a number of first semiconductor layers (which serve as sacrificial layers) and a number of second semiconductor layers (which serve as channel layers), in some embodiments. The method **200** continues to operation **206** of forming a dummy gate structure over the first to third fin structures. Next, the method **200** continues to operation **208** of recessing portions of each of the first to third fin structures. The method **200** then proceeds to operation **210** of recessing end portions of the first semiconductor layers of each of the first to third fin structures, thereby forming a first channel structure, a second channel structure, and a third channel structure, respectively. Next, the method **200** continues to operation **212** of forming a number of first inner spacers along sidewalls of the first channel structure, a number of second inner spacers along sidewalls of the second channel structure, and a number of third inner spacers along sidewalls of the third channel structure. Next, the method **200** proceeds to operation **214** of forming first source/drain structures along the sidewalls of the first channel structure, second source/drain structures along the sidewalls of the second channel structure, and third source/drain structures along the sidewalls of the third channel structure. The method **200** proceeds to operation **216** of removing a portion of the dummy gate structure and the second channel structure, while shaping sidewalls of the second inner spacers. Then, the method **200** continues to operation **218** of forming a first metal gate structure and a second metal gate structure.

[0019] As mentioned above, FIGS. **3-14C** each illustrate, in a cross-sectional view, a portion of a GAA FET device **300** at various fabrication stages of the method **200** of FIG. **2**. The GAA FET device **300** is similar to the GAA FET device **100** shown in FIG. **1**, but with certain features/structures/regions not shown, for the purposes of brevity. For example, the following figures of the GAA FET device **300** do not include source/drain structures (e.g., **110** of FIG. **1**). It should be understood the GAA FET device **300** may further include a number of other devices (not shown in the following figures) such as inductors, fuses, capacitors, coils, etc., while remaining within the scope of the present disclosure.

[0020] Corresponding to operation **202** of FIG. **2**, FIG. **3** is a cross-sectional view of the GAA FET device **300** including a semiconductor substrate **302** at one of the various stages of fabrication. The cross-sectional view of FIG. **3** is cut in a direction along the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section A-A indicated in FIG. **1**).

[0021] The substrate **302** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **302** may be a wafer, such as a silicon wafer. Generally, an SOI substrate includes a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **302** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

[0022] Corresponding to operation **204** of FIG. **2**, FIG. **4A** is a cross-sectional view of the GAA FET device **300** including a number of fin structures **401A**, **401B**, and **401C** formed over the substrate **302** at one of the various stages of fabrication. The cross-sectional view of FIG. **4A** is cut in a direction perpendicular to the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section B-B indicated in FIG. **1**). The fin structures **401A** to **401C** may sometimes be collectively referred to as fin structure **401** in the following discussions. Further,

FIG. 4B illustrates a top view of the GAA FET device **300** at the stage of fabrication that corresponds to operation **204**.

[0023] To form the fin structure **401**, a number of first semiconductor layers **410** and a number of second semiconductor layers **420** are alternately disposed (e.g., deposited) on top of one another to first form a stack. For example, one of the second semiconductor layers **420** is disposed over one of the first semiconductor layers **410** then another one of the first semiconductor layers **420** is disposed over the second semiconductor layer **410**, so on and so forth. The first and second stacks may include any number of alternately disposed first and second semiconductor layers **410** and **420**, respectively. For example in FIG. 4A, the stack includes 3 first semiconductor layers **410**, with 3 second semiconductor layers **420** alternately disposed therebetween and with one of the second semiconductor layers **420** being the topmost semiconductor layer. It should be understood that the GAA FET device **300** can include any number of first semiconductor layers and any number of second semiconductor layers, with either one of the first or second semiconductor layers being the topmost semiconductor layer, while remaining within the scope of the present disclosure.

[0024] The semiconductor layers **410** and **420** may have respective different thicknesses. Further, the first semiconductor layers **410** may have different thicknesses from one layer to another layer. The second semiconductor layers **420** may have different thicknesses from one layer to another layer. The thickness of each of the semiconductor layers **410** and **420** may range from few nanometers to few tens of nanometers. The first layer of the stack may be thicker than other semiconductor layers **410** and **420**. In an embodiment, each of the first semiconductor layers **410** has a thickness ranging from about 5 nanometers (nm) to about 20 nm, and each of the second semiconductor layers **420** has a thickness ranging from about 5 nm to about 20 nm.

[0025] The two semiconductor layers **410** and **420** have different compositions. In various embodiments, the two semiconductor layers **410** and **420** have compositions that provide for different oxidation rates and/or different etch selectivity between the layers. In an embodiment, the first semiconductor layers **410** include silicon germanium (Si.sub.1-xGe.sub.x), and the second semiconductor layers include silicon (Si). In an embodiment, each of the semiconductor layers **420** is silicon that may be undoped or substantially dopant-free (i.e., having an extrinsic dopant concentration from about 0 cm.sup.-3 to about 1×10.sup.17 cm.sup.-3), where for example, no intentional doping is performed when forming the layers **420** (e.g., of silicon).

[0026] In various embodiments, the semiconductor layers **420** may be intentionally doped. For example, when the GAA FET device **300** is configured as an n-type transistor (and operates in an enhancement mode), each of the semiconductor layers **420** may be silicon that is doped with a p-type dopant such as boron (B), aluminum (Al), indium (In), and gallium (Ga); and when the GAA FET device **300** is configured as a p-type transistor (and operates in an enhancement mode), each of the semiconductor layers **420** may be silicon that is doped with an n-type dopant such as phosphorus (P), arsenic (As), antimony (Sb). In another example, when the GAA FET device **300** is configured as an n-type transistor (and operates in a depletion mode), each of the semiconductor layers **420** may be silicon that is doped with an n-type dopant instead; and when the GAA FET device **300** is configured as a p-type transistor (and operates in a depletion mode), each of the semiconductor layers **420** may be silicon that is doped with a p-type dopant instead. In some embodiments, each of the semiconductor layers **410** is Si.sub.1-xGe.sub.x that includes less than 50% (x<0.5) Ge in molar ratio. For example, Ge may comprise about 15% to 35% of the semiconductor layers **410** of Si.sub.1-xGe.sub.x in molar ratio. Furthermore, the first semiconductor layers **410** may include different compositions among them, and the second semiconductor layers **420** may include different compositions among them.

[0027] Either of the semiconductor layers **410** and **420** may include other materials, for example, a compound semiconductor such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor such as GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, and/or GaInAsP, or combinations thereof. The materials of the

semiconductor layers **410** and **420** may be chosen based on providing differing oxidation rates and/or etch selectivity.

[0028] The semiconductor layers **410** and **420** can be epitaxially grown from the semiconductor substrate **302**. For example, each of the semiconductor layers **410** and **420** may be grown by a molecular beam epitaxy (MBE) process, a chemical vapor deposition (CVD) process such as a metal organic CVD (MOCVD) process, and/or other suitable epitaxial growth processes. During the epitaxial growth, the crystal structure of the semiconductor substrate **302** extends upwardly, resulting in the semiconductor layers **410** and **420** having the same crystal orientation with the semiconductor substrate **302**.

[0029] Upon growing the semiconductor layers **410** and **420** (e.g., as a stack) on the semiconductor substrate **302**, the stack may be patterned to form one or more fin structures (e.g., fin structures **401A** to **401C** shown in FIGS. **4A-B**). As further shown in the top view of FIG. **4B**, each of the fin structures **401A** to **401C** is elongated along a lateral direction (e.g., the X-direction). And as shown in FIG. **4A**, each of the fin structures **401A** to **401C** includes a stack of patterned semiconductor layers **410-420** interleaved with each other. The fin structures **401A** to **401C** are formed by patterning the stack of semiconductor layers **410-420** and the semiconductor substrate **302** using, for example, photolithography and etching techniques. Portions of the semiconductor layers **420** may collectively function as a conduction channel of the GAA FET device **300**, and the semiconductor layers **410** may be later removed and replaced with a portion of an active gate structure that wraps around each of the semiconductor layers **420**. Accordingly, the semiconductor layers **410** and **420** may sometimes be referred to as sacrificial layers **410** and channel layer **420**, respectively.

[0030] For example, a mask layer **402** (which can include multiple layers such as, for example, a pad oxide layer and an overlying hardmask layer) is formed over the topmost semiconductor layer (e.g., **420** in FIG. **4A**). The pad oxide layer may be a thin film comprising silicon oxide formed, for example, using a thermal oxidation process. The pad oxide layer may act as an adhesion layer between the topmost semiconductor layer **420** and the hardmask layer **402**. In some embodiments, the hardmask layer may include silicon nitride, silicon oxynitride, silicon carbonitride, the like, or combinations thereof. In some other embodiments, the hardmask layer may include a material similar as a material of the semiconductor layers **410/420** such as, for example, Si.sub.1-yGe.sub.y, Si, etc., in which the molar ratio (y) may be different from or similar to the molar ratio (x) of the semiconductor layers **410**. The hardmask layer may be formed over the stack (i.e., before patterning the stack) using low-pressure chemical vapor deposition (LPCVD) or plasma enhanced chemical vapor deposition (PECVD), for example.

[0031] The mask layer may be patterned using photolithography techniques. Generally, photolithography techniques utilize a photoresist material (not shown) that is deposited, irradiated (exposed), and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material, such as the mask layer in this example, from subsequent processing steps, such as etching. For example, the photoresist material is used to pattern the pad oxide layer and pad nitride layer to form a patterned mask.

[0032] The patterned mask can be subsequently used to pattern exposed portions of the semiconductor layers **410-420** and the substrate **302** to form one or more of the fin structures **410**, thereby defining trenches **430** (or openings) between adjacent fin structures **401**. When multiple fin structures are formed, such a trench **430** may be disposed between any adjacent ones of the fin structures **401**. In some embodiments, the fin structure **401** is formed by etching trenches in the semiconductor layers **410-420** and substrate **302** using, for example, reactive ion etch (RIE), neutral beam etch (NBE), the like, or combinations thereof. The etch may be anisotropic. In some embodiments, the trenches may be strips (when viewed from the top) parallel to each other, and closely spaced with respect to each other. In some embodiments, the trenches may be continuous and surround the fin structure **401**.

[0033] Corresponding to operation **206** of FIG. 2, FIG. 5A is a cross-sectional view of the GAA FET device **300** including a dummy gate structure **500**, at one of the various stages of fabrication. The cross-sectional view of FIG. 5A is cut in a direction along the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section A-A indicated in FIG. 1). Further, FIG. 5B illustrates a top view of the GAA FET device **300** at the stage of fabrication that corresponds to operation **206**.

[0034] The dummy gate structure **500** may have a lengthwise direction perpendicular to the lengthwise direction of the fin structure **401**, as better seen in FIG. 5B. As such, the dummy gate structure **500** may be formed to overlay (e.g., straddle) a portion of the fin structure **401**. For example, the dummy gate structure **500** may straddle a central portion of the fin structure **401**, such that end or side portions of the fin structure **401** are exposed, as shown in FIGS. 5A-B.

[0035] In some embodiments, the dummy gate structure **500** may contact the top surface of an isolation structure (not shown) embedding a lower portion of the fin structure **401**, with its bottom surface. Such an isolation structure, typically referred to as a shallow trench isolation (STI), includes a number of portions, each of which is disposed between neighboring fin structures. In some other embodiments, the neighboring fin structures may be further separated by a cladding layer formed over the STI, which will be removed in a later fabrication stage. In such embodiments, the bottom surface of the dummy gate structure **500** may not contact the top surface of the STI.

[0036] The dummy gate structure **500** may include a dummy gate dielectric and a dummy gate, which are not shown separately for purpose of clarity. To form the dummy gate structure **500**, a dielectric layer may be formed over the fin structure **401**. The dielectric layer may be, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, silicon oxycarbide, multilayers thereof, or the like, and may be deposited or thermally grown.

[0037] A gate layer is formed over the dielectric layer, and a mask layer is formed over the gate layer. The gate layer may be deposited over the dielectric layer and then planarized, such as by a CMP. The mask layer may be deposited over the gate layer. The gate layer may be formed of, for example, polysilicon, although other materials may also be used. The mask layer may be formed of, for example, silicon nitride or the like.

[0038] After the layers (e.g., the dielectric layer, the gate layer, and the mask layer) are formed, the mask layer may be patterned using suitable lithography and etching techniques. Next, the pattern of the mask layer may be transferred to the gate layer and the dielectric layer by a suitable etching technique to form the dummy gate structure **500**.

[0039] Upon forming the dummy gate structure **500**, a gate spacer **502** may be formed on opposing sidewalls of the dummy gate structure **500**, as shown in FIGS. 5A-B. The gate spacer **502** may each be a low-k spacer and may be formed of a suitable dielectric material, such as silicon oxide, silicon nitride, silicon carbide, silicon carbonitride, silicon oxynitride, silicon oxycarbonitride, or the like. Any suitable deposition method, such as thermal oxidation, chemical vapor deposition (CVD), or the like, may be used to form the gate spacer **502**. The shapes and formation methods of the gate spacer **502**, as illustrated in FIGS. 5A-B, are merely non-limiting examples, and other shapes and formation methods are possible. These and other variations are fully intended to be included within the scope of the present disclosure.

[0040] Corresponding to operation **208** of FIG. 2, FIG. 6 is a cross-sectional view of the GAA FET device **300** in which (e.g., end) portions of the fin structure **401** that are not overlaid by the dummy gate structure **500** (together with the gate spacer **502**) are recessed, at one of the various stages of fabrication. The cross-sectional view of FIG. 6 is cut in a direction along the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section A-A indicated in FIG. 1).

[0041] As shown, the dummy gate structure **500** (together with the gate spacer **502**) can serve as a

mask to recess (e.g., etch) the non-overlaid portions of the fin structure **401**, which results in the remaining fin structure **401** having respective remaining portions of the semiconductor layers **410** and **420** alternately stacked on top of one another. As a result, recesses **602** can be formed on opposite sides of the remaining fin structure **410**. A source/drain structure can be formed in each of the recesses **602** in a later fabrication stage. Accordingly, the recess **602** is sometimes referred to as a source/drain (S/D) recess. The recesses **602** can expose respective sidewalls of the remaining semiconductor layers **410** and **420**. In some embodiments, such newly exposed sidewalls of the semiconductor layers **410** and **420** may vertically align with a sidewall of the gate spacer **502**. Alternatively stated, a single contact face, consisting of the sidewalls of the semiconductor layers **410** and **410**, can be formed on each side of the dummy gate structure **500**.

[0042] The recessing step to form the recesses **602** may be configured to have at least some anisotropic etching characteristic. For example, the recessing step can include a plasma etching process, which can have a certain amount of anisotropic characteristic. In such a plasma etching process (including radical plasma etching, remote plasma etching, and other suitable plasma etching processes), gas sources such as chlorine (Cl.sub.2), hydrogen bromide (HBr), carbon tetrafluoride (CF.sub.4), fluoroform (CHF.sub.3), difluoromethane (CH.sub.2F.sub.2), fluoromethane (CH.sub.3F), hexafluoro-1,3-butadiene (C.sub.4F.sub.6), boron trichloride (BCl.sub.3), sulfur hexafluoride (SF.sub.6), hydrogen (H.sub.2), nitrogen trifluoride (NF.sub.3), and other suitable gas sources and combinations thereof can be used with passivation gases such as nitrogen (N.sub.2), oxygen (O.sub.2), carbon dioxide (CO.sub.2), sulfur dioxide (SO.sub.2), carbon monoxide (CO), methane (CH.sub.4), silicon tetrachloride (SiCl.sub.4), and other suitable passivation gases and combinations thereof. Moreover, for the recessing step, the gas sources and/or the passivation gases can be diluted with gases such as argon (Ar), helium (He), neon (Ne), and other suitable dilutive gases and combinations thereof.

[0043] Corresponding to operation **210** of FIG. 2, FIG. 7A is a cross-sectional view of the GAA FET device **300** in which respective end portions of the semiconductor layers **410** are recessed, at one of the various stages of fabrication. In FIG. 7B, illustrated is a cross-sectional view of another embodiment of the GAA FET device **300**. The cross-sectional views of FIGS. 7A and 7B are each cut in a direction along the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section A-A indicated in FIG. 1).

[0044] The end portions of the semiconductor layers **410** can be removed (e.g., etched) using a “pull-back” process to pull the semiconductor layers **410** back by a pull-back distance. In an example where the semiconductor layers **420** include Si, and the semiconductor layers **410** include SiGe, the pull-back process may include a hydrogen chloride (HCl) gas isotropic etch process, which etches SiGe without attacking Si. As such, the Si layers (nanostructures) **420** may remain substantially intact during this process. Consequently, a pair of recesses, **702**, can be formed on the ends of each semiconductor layer **410**, with respect to the neighboring semiconductor layers **420**. Alternatively or additionally, while etching the semiconductor layers **410**, portions of the semiconductor layers **420** may also be etched back by a pull-back distance, which is generally controlled to be close to zero by selecting an etchant that has a high etching selectivity between the materials of the semiconductor layers **410** and **420**.

[0045] Referring to FIGS. 7A and 7B, the gate spacer **502** may remain substantially intact during the pull-back process of the semiconductor layers **410**. In such embodiments, the etchant used in the pull-back process can have a high etching selectivity between the materials of the gate spacer **502** and the semiconductor layers **410**. For example, the etchant (e.g., Cl.sub.2, HBr, CF.sub.4, CHF.sub.3, CH.sub.2F.sub.2, or CH.sub.3F) may etch the semiconductor layers **410** at a substantially higher rate than etch the gate spacer **502**. As such, a top surface of the topmost semiconductor layer of the fin structure **401** (e.g., **420**) may remain covered by the dummy gate structure **500** and the gate spacer **502**.

[0046] In the embodiment of FIG. 7A, the recess **702** can be formed with a straight-line-based

sidewall. Accordingly, each recess **702** can form a pair of right angles, one of which is between the sidewall of a corresponding semiconductor layer **410** and the bottom surface of an upper neighboring semiconductor layer **420**, and the other of which is between the sidewall of the corresponding semiconductor layer **410** and the top surface of a lower neighboring semiconductor layer **420**. In the embodiments of FIG. 7B, the recess **702** can be formed with a curvature-based sidewall. The curvature-based sidewall may inwardly protrude toward the corresponding semiconductor layer **410**. Such a curvature-based sidewall may be achieved by increasing an isotropic etching characteristic of the pull-back process, such that the semiconductor layers **410** may be etched non-uniformly along its sidewall.

[0047] Corresponding to operation **212** of FIG. 2, FIG. 8 is a cross-sectional view of the GAA FET device **300** including a number of inner spacers **802**, at one of the various stages of fabrication. The cross-sectional view of FIG. 8 is cut in a direction along the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section A-A indicated in FIG. 1). The inner spacers **802** shown in FIG. 8 are formed based on the recesses **702** shown in FIG. 7A, and thus, it should be appreciated that the inner spacers **802** can have different profiles than FIG. 8 (e.g., based on the recesses **702** of FIG. 7B), while remaining within the scope of the present disclosure.

[0048] The inner spacers **802** can be formed conformally by chemical vapor deposition (CVD), or by monolayer doping (MLD) of nitride followed by spacer RIE. The inner spacers **802** can be deposited using, e.g., a conformal deposition process and subsequent isotropic or anisotropic etch back to remove excess spacer material on the sidewalls of the fin structures **401** and on a surface of the semiconductor substrate **302** (as shown in FIG. 8). A material of the inner spacers **802** can be formed from the same or different material as the gate spacer **502** (e.g., silicon nitride). For example, the inner spacers **802** can be formed of silicon nitride (SiN), silicoboron carbonitride (SiBCN), silicon carbonitride (SiCN), silicon carbon oxynitride (SiON), silicon oxycarbonitride (SiOCN), or any other type of dielectric material (e.g., a dielectric material having a dielectric constant less than about 5) appropriate to the role of forming an insulating gate sidewall spacers of transistors.

[0049] Corresponding to operation **214** of FIG. 2, FIG. 9A is a cross-sectional view of the GAA FET device **300** including source/drain structures **902** formed over the workpiece shown in FIG. 8, at one of the various stages of fabrication. The cross-sectional view of FIG. 9A is cut in a direction along the lengthwise direction of a number of channel layers of the GAA FET device **300** (e.g., cross-section A-A indicated in FIG. 1). Further, FIG. 9B illustrates a top view of the GAA FET device **300** at the stage of fabrication that corresponds to operation **214**, in which the source/drain structures **902** are just formed.

[0050] The source/drain structure **902** is disposed in a corresponding one of the S/D recesses **602**. As shown in FIGS. 9A-B, the source/drain structures **902** are disposed on the opposite sides of each of the fin structures **401A** to **401C**. The source/drain structures **902** are coupled to the semiconductor (channel) layers **420**, and separated from the semiconductor (sacrificial) layers **410** with the inner spacer **802** disposed therebetween. The source/drain structures **902** are formed by epitaxially growing a semiconductor material in the recesses **602** using suitable methods such as metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), vapor phase epitaxy (VPE), selective epitaxial growth (SEG), the like, or combinations thereof.

[0051] Upon forming the source/drain structures **902**, an interlayer dielectric (ILD) **904** is formed over the source/drain structure **902**. The ILD **904** is formed of a dielectric material such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), undoped silicate glass (USG), or the like, and may be deposited by any suitable method, such as CVD, PECVD, or FCVD. After the ILD is formed, an optional dielectric layer (not shown) is formed over the ILD. The dielectric layer can function as a protection layer to prevent or reduces the loss of the ILD in subsequent etching processes. The dielectric layer may be formed of a suitable material, such as silicon nitride, silicon carbonitride, or the like, using a suitable method

such as CVD, PECVD, or FCVD. After the dielectric layer is formed, a planarization process, such as a CMP process, may be performed to achieve a level top surface for the dielectric layer. After the planarization process, the top surface of the dielectric layer is level with the top surface of the dummy gate structure **500**, in some embodiments.

[0052] Corresponding to operation **216** of FIG. **2**, FIG. **10A** is a top view of the GAA FET device **300** in which one of the fin structures **401A** to **401C** is removed, at one of the various stages of fabrication. Further, FIGS. **10B** and **10C** each illustrate a cross-sectional view of the GAA FET device **300** at the stage of fabrication that corresponds to operation **216**. The cross-sectional view of FIG. **10B** is cut in a direction along the lengthwise direction of one of the channel structures that remains, e.g., **401A**; and the cross-sectional view of FIG. **10C** is cut in a direction along the lengthwise direction of one of the channel structures that has been partially removed, e.g., **401B**.

[0053] Subsequently to forming the ILD **904** (not shown in FIG. **10A**), a portion of the dummy gate structure **500** and a majority portion of one of the channel structures **401A** to **401C** overlaid by the portion of the dummy gate structure **500** is removed to form a cavity. The cavity can (e.g., vertically) extend into the substrate **302**. Such a cavity can be filled with a dielectric material to form the isolation structure **1000**. In some embodiments, the dielectric material may be a Si-based material such as, for example, SiO, SiCN, SiN, SiON, or combinations thereof. In some other embodiments, the dielectric material may be a metal-based material such as, for example, Al.sub.2O.sub.3. The isolation structure **1000** can thus separate remaining portions of the dummy gate structure **500** from each other.

[0054] For example in FIG. **10A**, the dummy gate structure **500** can be grouped into three portions, **500A**, **500B**, and **500C**, which overlay the channel structures **401A**, **401B**, and **401C**, respectively. Upon forming the ILD **904**, the portion **500B** may be first removed, which exposes the channel structure **401B**. Next, a majority portion of the channel structure **401B**, which includes its remaining sacrificial layers **410** and a central portion of each of the channel layers **420**, is removed until a cavity **1001** extends into the substrate **302** with a certain depth (FIG. **10C**). Next, the cavity **1001** is filled with a dielectric material thereby forming the isolation structure **1000**. As a result, the isolation structure **1000** is interposed between the source/drain structures **902** that were coupled to the removed channel structure **401B**, as shown in FIGS. **10A** and **10C**, while the remaining channel structure **401A** (and **401C**) is still overlaid by the corresponding portion of the dummy gate structure **500A** and coupled to its source/drain structures **902**, as shown in FIGS. **10A** and **10B**. Specifically in FIG. **10C**, the central portions of the channel layers **420** (together with the remaining sacrificial layers **410** and the portion of the dummy gate structure **500B**) are replaced with the isolation structure **1000**, while end portions of the channel layers **420** remain.

[0055] In various embodiments of the present disclosure, the removal of the portion of the dummy gate structure **500B** and the majority portion of the channel structure **401B** may be concurrently or respectively performed. For example, when the dummy gate structure **500** is formed of similar material(s) to the semiconductor (channel and sacrificial) layers of the channel structure **401B**, e.g., Si, SiGe, etc., the portion of the dummy gate structure **500B** and the majority portion of the channel structure **401B** can be concurrently removed through one or more (e.g., plasma) etching processes. In such a plasma etching process (including radical plasma etching, remote plasma etching, and other suitable plasma etching processes), gas sources such as chlorine (Cl.sub.2), carbon tetrafluoride (CF.sub.4), fluoroform (CHF.sub.3), difluoromethane (CH.sub.2F.sub.2), fluoromethane (CH.sub.3F), and other suitable gas sources and combinations thereof can be used with passivation gases such as nitrogen (N.sub.2), oxygen (O.sub.2), carbon dioxide (CO.sub.2), sulfur dioxide (SO.sub.2), carbon monoxide (CO), methane (CH.sub.4), silicon tetrachloride (SiCl.sub.4), and other suitable passivation gases and combinations thereof. The gas sources and/or the passivation gases can be diluted with gases such as argon (Ar), helium (He), neon (Ne), and other suitable dilutive gases and combinations thereof.

[0056] Further, concurrently with removing the portion of the dummy gate structure **500B** and the

majority portion of the channel structure **401B**, the etching process can be tuned to have less selectivity to the dielectric material that the inner spacers **802** contain. For example, a flow rate of one or more of the etchant gases, which has a less etching selectivity with respect to the dielectric material of the inner spacers **802** (SiN/SiCN/SiOCN/SiON), such as Cl.sub.2, CF.sub.4, CHF.sub.3, CH.sub.2F.sub.2, and CH.sub.3F may be increased. Stated another way, the inner spacers **802** may also be etched when removing the portion of the dummy gate structure **500B** and the majority portion of the channel structure **401B**. As such, sidewalls of each of the inner spacers **802**, exposed in the cavity **1001**, can be shaped or otherwise patterned. This allows the inwardly protruding sidewall of an inner spacer **802** (with respect to its corresponding sacrificial layer **410**) to be removed, thereby causing the inner spacer **802** to have a nearly vertical sidewall facing the cavity **1001**. With this nearly vertical profile, the etchant gas used to form the cavity may not be blocked, which allows the etchant gas to reach a lower portion of the to-be-removed channel structure **401B**. Consequently, the isolation structure **1000**, inheriting the profile of the cavity **1001**, can be successfully formed to separate the dummy gate portions **500A** and **500C**, which will later be replaced with metal gate structures, respectively.

[0057] FIGS. **11A-11L** illustrate various profiles of the inner spacer **802** with one of its sidewalls contacting the isolation structure **1000**, in accordance with various embodiments. For example, profiles on the left-hand side (boxed with dotted lines) are a first sidewall of the inner spacer **802** contacting the source/drain structure **902**, and profiles on the right-hand side are a second sidewall of the inner spacer **802** contacting the isolation structure **1000**. Accordingly, a first interface is present between the first sidewall of the inner spacer **802** and the source/drain structure **902**, and a second interface is present between the second sidewall of the inner spacer **802** and the isolation structure **1000**.

[0058] In some embodiments, as shown in FIGS. **11A-C** and **11G-I**, the inner spacer **802** may be formed of a single film. In some embodiments, as shown in FIGS. **11D-F** and **11J-L**, the inner spacer **802** may be formed of multiple films. Further, in some embodiments, as shown in FIGS. **11G-L**, the inner spacer **802** may include a void or seam **1102**. In some embodiments, as shown in FIGS. **11A**, **11D**, **11G**, and **11J**, profiles of the first sidewall of the inner spacer **802** facing the source/drain structure **902** may be substantially straight, while, as shown in FIGS. **11B**, **11C**, **11E**, **11F**, **11H**, **11I**, **11K** and **11L**, profiles of the first sidewall of the inner spacer **802** may have a shape concave toward the inner spacer itself. In some embodiments, as shown in FIGS. **11A**, **11B**, **11D**, **11E**, **11G**, **11H**, **11J**, and **11K**, profiles of the second sidewall of the inner spacer **802** facing the isolation structure **1000** may have a shape concave toward the isolation structure **1000**. In some embodiments, as shown in FIGS. **11C**, **11F**, **11I**, and **11L**, profiles of the second sidewall of the inner spacer **802** facing the isolation structure **1000** may be substantially straight.

[0059] In various embodiments, one or more profiles of the inner spacer **802** contacting the isolation structure **1000** (used to contact the removed fin structure **401B**) can be characterized with a number of critical dimensions. Similarly, one or more profiles of the inner spacer **802** contacting the remaining channel structure (e.g., **401A** and **401C**) can also be characterized with a number of critical dimensions. For illustration purposes, the inner spacer **802** contacting the isolation structure **1000** and the inner spacer **802** contacting the remaining channel structure are herein referred to as “inner spacer **802I**” and “inner spacer **802C**,” respectively. According the various embodiments, the inner spacer **802I** has a first interface present between one of its sidewalls and a corresponding source/drain structure, and a second interface present between the other of its sidewalls and a corresponding isolation structure; and the inner spacer **802C** has a first interface present between one of its sidewalls and a corresponding source/drain structure, and a second interface present between the other of its sidewalls and a corresponding channel structure (e.g., a channel layer).

[0060] For example in FIG. **12A**, a profile of the inner spacer **802C** is shown, in comparison with different profiles of the inner spacer **802I** shown in FIGS. **12B** and **12C**, respectively. Further, the profiles shown in FIGS. **12B** and **12C** refer to different amounts of the inner spacer **802I** being

etched (in operation 212 of FIG. 2), in some embodiments. Similarly, FIG. 12D illustrates a profile of the inner spacer 802C, in comparison with different profiles of the inner spacer 802I shown in FIGS. 12E and 12F that refer to different amounts of the inner spacer 802I being etched, respectively; and FIG. 12G illustrates a profile of the inner spacer 802C, in comparison with different profiles of the inner spacer 802I shown in FIGS. 12H and 12I that refer to different amounts of the inner spacer 802I being etched, respectively.

[0061] As illustrated in FIG. 12A (which serves as a representative example for the inner spacer 802C), the inner spacer 802C is characterized by critical dimensions, $CD.sub.x'$, $CD.sub.y'$, and $\theta.sub.O$; and as illustrated in FIG. 12B (which serves as a representative example for the inner spacer 802I), the inner spacer 802I is characterized by critical dimensions, $CD.sub.x$, $CD.sub.y$, and $\theta.sub.C$.

[0062] Specifically in FIG. 12A, $CD.sub.x'$ represents a closest distance laterally measured from the tip of a seam within the inner spacer 802C to a sidewall of the inner spacer 802C facing a corresponding channel structure; $CD.sub.y'$ represents a distance vertically measured from the seam to a top or bottom surface of the inner spacer 802C; and $\theta.sub.O$ represents an angle between the sidewall of the inner spacer 802C and the top or bottom surface of the inner spacer 802C.

[0063] Specifically in FIGS. 12B and 12C, $CD.sub.x$ represents a closest distance laterally measured from the tip of a seam within the inner spacer 802I to a sidewall of the inner spacer 802I facing a corresponding isolation structure; $CD.sub.y$ represents a distance vertically measured from the seam to a top or bottom surface of the inner spacer 802I; and $\theta.sub.C$ represents an angle between the sidewall of the inner spacer 802I and the top or bottom surface (or its projection) of the inner spacer 802I. Further, for the inner spacer 802I (e.g., FIG. 12B), an enlarged profile of a connecting portion 1201 disposed between the nearly vertical sidewall and the top/bottom surface is shown in FIG. 12J. With this connecting portion present, the inner spacer 802I can be further characterized by a critical dimension $\theta.sub.R$ that represents an angle between the top/bottom surface and the connecting portion of the inner spacer 802I.

[0064] In accordance with various embodiments of the present disclosure, one of the sidewalls of the inner spacer 802I is exposed and patterned during operation 216, while the inner spacer 802C may remain substantially intact during operation 216 with both of the sidewalls kept covered. Accordingly, $CD.sub.x'$ can be substantially greater than $CD.sub.x$, and $CD.sub.x$ can become substantially smaller as a patterned (e.g., etched) amount of the sidewall increases, as illustrated in the examples of FIGS. 12A-C, 12D-F, and 12G-I, respectively.

[0065] In the example of FIGS. 12A-C, the inner spacer 802C and inner spacer 802I (before being patterned) may both present a profile having one sidewall (facing a corresponding source/drain structure) with a straight profile and the other sidewall (facing a corresponding channel structure) with a protruding profile, e.g., similar to FIG. 11G. Following the patterning step (operation 216), the protruding sidewall of the inner spacer 802C may remain, while the protruding sidewall of the inner spacer 802I may be patterned to have a straight profile along its majority portion.

[0066] In the example of FIGS. 12D-F, the inner spacer 802C and inner spacer 802I (before being patterned) may both present a profile having one sidewall (facing a corresponding source/drain structure) with a concave profile and the other sidewall (facing a corresponding channel structure) with a protruding profile, e.g., similar to FIG. 11H. Following the patterning step (operation 216), the protruding sidewall of the inner spacer 802C may remain, while the protruding sidewall of the inner spacer 802I may be patterned to have a straight profile along its majority portion.

[0067] In the example of FIGS. 12G-I, the inner spacer 802C and inner spacer 802I (before being patterned) may both present a profile having one sidewall (facing a corresponding source/drain structure) with a concave profile and the other sidewall (facing a corresponding channel structure) with a straight profile, e.g., similar to FIG. 11I. Following the patterning step (operation 216), the straight sidewall of the inner spacer 802C may remain, while the straight sidewall of the inner spacer 802I may be patterned to retreat toward its concave sidewall.

[0068] In some embodiments where the examples of FIGS. 12A-C, FIGS. 12D-F, and FIGS. 12G-I are respectively applied, since the inner spacer **802C** generally have a sidewall (facing the channel structure) that laterally extends with a more distance than **802I**, $CD.sub.x'-CD.sub.x$ may be equal to or greater than about 0.3 nanometers (nm), where $CD.sub.x'$ is in a range from about 2 nm to about 30 nm, and $CD.sub.x$ is in a range from about 0 nm to about 29.7 nm. It should be noted that the scenario where $CD.sub.x'=0$ (FIGS. 12C, 12F, and 12I) may occur when the seam tip reaches the sidewall of the inner spacer **802I**, in some embodiments. In some embodiments, $CD.sub.y'$ may be substantially close to $CD.sub.y$ (i.e., with a ratio ranging from about 0.9 to about 1.1), where $CD.sub.y'$ is in a range from about 2 nm to about 30 nm, and $CD.sub.y$ is in a range from about 2 nm to about 30 nm.

[0069] Further, in the examples where the protruding sidewall of the inner spacer **802C** remains (FIGS. 12A and 12D), $\theta.sub.O$ may be in a range of about 30° to about 87° . In comparison, with a majority portion of the protruding sidewall of the inner spacer **802I** being etched, $\theta.sub.C$ may be in a range of about 75° to about 135° . When $\theta.sub.C$ is equal to 90° , it refers to the scenario where the “patterned” sidewall of the inner spacer **802I** has a nearly vertical profile. According to various embodiments of the present disclosure, $\theta.sub.C$ should not be less than 90° with more than 15° , and should be equal to or greater than 90° . With its majority portion being patterned as a straight profile, the sidewall of the inner spacer **802I** can have a connecting portion between the nearly straight sidewall and its top/bottom surface, as illustrated in the enlarged view of FIG. 12J. As shown, $\theta.sub.R$ may be an acute angle, which is in a range of about 30° to about 87° . In the example where the sidewall of the inner spacer **802C** is formed as a straight profile (FIG. 12C), $\theta.sub.O$ may be close to about 90° . Similarly, with the already nearly vertical sidewall of the inner spacer **802I** being retreated, $\theta.sub.C$ may still be close to about 90° .

[0070] FIGS. 13A-E illustrate various profiles of the multi-layer inner spacer **802** (including the inner spacer **802C** and inner spacer **802I**) after operation **216**, in accordance with various embodiments. In the examples of FIGS. 13A-E, each of the inner spacer **802C** and inner spacer **802I** is formed to have a first layer and a second layer (i.e., before performing operation **216**), where both the inner spacer **802C** and inner spacer **802I** may have their sidewalls (facing a corresponding channel structure) presenting a protruding profile, and their other sidewalls (facing a corresponding source/drain structure) presenting a nearly vertical profile. It should be appreciated that such a multi-layer inner spacer **802** can also have their sidewalls presenting different profiles than FIGS. 13A-E (e.g., FIGS. 11K, 11L, etc.), while remaining within the scope of the present disclosure.

[0071] In some embodiments, the first layer may be first deposited, followed by deposition of the second layer. As such, the first layer can line the inner spacer recess (e.g., **702**), with the rest of the inner spacer recess filled with the second layer. Upon operation **216** being performed, a majority portion of the protruding sidewall of the inner spacer **802I**, which may be formed by the first layer or even the second layer, may be patterned to have a nearly vertical profile. The profile of each of the inner spacer **802C** and inner spacer **802I** can be characterized with a respective number of critical dimensions.

[0072] As illustrated in FIG. 13A (which serves as a representative example for the inner spacer **802C**), the inner spacer **802C** is characterized by critical dimensions, $CD.sub.x1'$, $CD.sub.x2'$, $CD.sub.y1'$, $CD.sub.y2'$, and $\theta.sub.O$; and as illustrated in FIG. 13B (which serves as a representative example for the inner spacer **802I**), the inner spacer **802I** is characterized by critical dimensions, $CD.sub.x1$, $CD.sub.x2$, $CD.sub.y1$, $CD.sub.y2$, $\theta.sub.O$, and $\theta.sub.R$.

[0073] Specifically in FIG. 13A, $CD.sub.x1'$ represents a distance laterally measured from a tip of the second layer of the inner spacer **802C** to a sidewall of the inner spacer **802C** facing a corresponding channel structure (alternatively, a laterally measured thickness of the first layer of the inner spacer **802C**); $CD.sub.x2'$: represents a closest distance laterally measured from the tip of a seam within the inner spacer **802C** to the tip of the second layer; $CD.sub.y1'$ represents a distance

vertically measured from a top or bottom surface of the second layer to a top or bottom surface of the inner spacer **802C** (alternatively, a vertically measured thickness of the first layer of the inner spacer **802C**); $CD.sub.y2'$ represents a distance vertically measured from the seam to the top or bottom surface of the second layer; and $\theta.sub.O$ represents an angle between the sidewall of the inner spacer **802C** and the top or bottom surface of the inner spacer **802C**.

[0074] Specifically in FIG. **13B**, $CD.sub.x1$ represents a distance laterally measured from a tip of the second layer of the inner spacer **802I** to a sidewall of the inner spacer **802I** facing a corresponding isolation structure (alternatively, a laterally measured thickness of the first layer of the inner spacer **802I**); $CD.sub.x2$ represents a closest distance laterally measured from the tip of a seam within the inner spacer **802I** to the tip of the second layer; $CD.sub.y1$ represents a distance vertically measured from a top or bottom surface of the second layer to a top or bottom surface of the inner spacer **802I** (alternatively, a vertically measured thickness of the first layer of the inner spacer **802I**); $CD.sub.y2$ represents a distance vertically measured from the seam to the top or bottom surface of the second layer; and $\theta.sub.C$ represents an angle between the sidewall of the inner spacer **802I** and the top or bottom surface (or its projection) of the inner spacer **802I**.

Similarly, with a connecting portion between the sidewall and the top/bottom surface present, the inner spacer **802I** can be further characterized by a critical dimension $\theta.sub.R$ that represents an angle between the top/bottom surface and the connecting portion of the inner spacer **802I**. It should be appreciated that $\theta.sub.O$, $\theta.sub.C$, and $\theta.sub.R$ may follow the trend similar to the trend when the inner spacers **802I** and **802C** are each formed of a single layer, and thus, the associated description will not be repeated.

[0075] In accordance with various embodiments of the present disclosure, one of the sidewalls of the inner spacer **802I** is exposed and patterned during operation **216**, while the inner spacer **802C** may remain substantially intact during operation **216** with both of the sidewalls kept covered. Accordingly, $CD.sub.x1'$ can be substantially greater than $CD.sub.x1$, and $CD.sub.x1$ (even $CD.sub.x2$) can become substantially smaller as a patterned (e.g., etched) amount of the sidewall increases, as illustrated in the example of FIGS. **13B-E**.

[0076] Starting with the example shown in FIGS. **13A-B**, the inner spacer **802C** and inner spacer **802I** (before being patterned) may both present a profile having one sidewall (facing a corresponding source/drain structure) with a straight profile and the other sidewall (facing a corresponding channel structure) with a protruding profile, e.g., similar to FIG. **11G**. Following the patterning step (operation **216**), the protruding sidewall of the inner spacer **802C** may remain, while the protruding sidewall of the inner spacer **802I** may be patterned to have a straight profile along its majority portion. As such, $CD.sub.x1'-CD.sub.x1$ may be equal to or greater than about 0.3 nanometers (nm). In some embodiments, $CD.sub.y1'$ may be substantially close to $CD.sub.y1$ and $CD.sub.y2'$ may be substantially close to $CD.sub.y2$ (i.e., with a ratio ranging from about 0.9 to about 1.1).

[0077] Continuing with the example shown in FIGS. **13B-E**, with an increasing amount of the protruding sidewall being etched, $CD.sub.x1$ reduces, and even $CD.sub.x2$ also reduces, which may affect at least one of: a difference between $CD.sub.x1$ and $CD.sub.y1$, a difference between $CD.sub.x2$ and $CD.sub.y2$, a difference between $CD.sub.y1'$ and $CD.sub.x1'$, or a difference between $CD.sub.y2'$ and $CD.sub.x2'$. In FIG. **13B** (where only the first layer is etched during operation **216**), $CD.sub.x1'-CD.sub.x1 \geq 0.3$ nm; $CD.sub.x2 \approx CD.sub.x2'$; $CD.sub.y1 \approx CD.sub.x1$; $CD.sub.y1-CD.sub.x1 \geq 0.3$ nm; $CD.sub.y1'-CD.sub.x1 \geq CD.sub.y1'-CD.sub.x1'$ with a difference being equal to or greater than 0.3 nm; and $CD.sub.y2 \approx CD.sub.x2$. In FIGS. **13C-D** (where both the first and second layers are etched during operation **216**), $CD.sub.x1'-CD.sub.x1 \geq 0.3$ nm; $CD.sub.x1=0$; $CD.sub.x2 < CD.sub.x2'$; $CD.sub.x2'-CD.sub.x2 \geq 0.3$ nm; $CD.sub.y1 > CD.sub.x1$; $CD.sub.y1-CD.sub.x1 \geq 0.3$ nm; $CD.sub.y1-CD.sub.x1 \geq CD.sub.y1-CD.sub.x1'$ with a difference being equal to or greater than 0.3 nm; $CD.sub.x2 < CD.sub.y2'$; $CD.sub.y2-CD.sub.x2 \geq 0.3$ nm; and $CD.sub.y2-CD.sub.x2 > CD.sub.y2'-CD.sub.x2'$; with a difference being equal to or greater than 0.3

nm. In FIG. 13E (where both the first and second layers are etched and the sidewall of the inner spacer **802I** reaches the tip of the seam during operation **216**), $CD.sub.x1'-CD.sub.x1 \geq 0.3$ nm; $CD.sub.x1=0$; $CD.sub.x2' < CD.sub.x2$; $CD.sub.x2'-CD.sub.x2 \geq 0.3$ nm; $CD.sub.x2=0$; $CD.sub.y1 > CD.sub.x1$; $CD.sub.y1-CD.sub.x1 \geq 0.3$ nm; $CD.sub.y1-CD.sub.x1 \geq CD.sub.y1'-CD.sub.x1'$ with a difference being equal to or greater than 0.3 nm; $CD.sub.x2 < CD.sub.y2$; $CD.sub.y2-CD.sub.x2 \geq 0.3$ nm; and $CD.sub.y2-CD.sub.x2 > CD.sub.y2'-CD.sub.x2'$ with a difference being equal to or greater than 0.3 nm.

[0078] In brief summary, when the last deposited layer of the inner spacer **802I** is not etched during operation **216** (e.g., FIG. 13B), $CD.sub.yn \approx CD.sub.xn$, where $CD.sub.yn$ represents a vertical distance between a seam within the inner spacer **802I** and a top/bottom surface of the last deposited layer, and $CD.sub.xn$ represents a lateral distance between a tip of the seam and a tip of the last deposited layer of the inner spacer **802I**. When a portion of the last deposited layer of the inner spacer **802I** is etched during operation **216** (e.g., FIGS. 13C and 13D), $CD.sub.x1=0$. When a portion of the last deposited layer of the inner spacer **802I** is etched and the sidewall of the inner spacer **802I** reaches the tip of the seam contained therein during operation **216** (e.g., FIG. 13E), $CD.sub.yn-CD.sub.xn \geq 0.3$ nm, and $CD.sub.yn-CD.sub.xn > CD.sub.yn'-CD.sub.xn'$ with a difference being equal to or greater than 0.3 nm, where $CD.sub.yn'$ represents a vertical distance between a seam within the inner spacer **802C** and a top/bottom surface of the last deposited layer, and $CD.sub.xn'$ represents a lateral distance between a tip of the seam and a tip of the last deposited layer of the inner spacer **802C**.

[0079] Corresponding to operation **218** of FIG. 2, FIG. 14A is a top view of the GAA FET device **300** including active (e.g., metal) gate structures **1400A** and **1400C**, at one of the various stages of fabrication. Further, FIGS. 14B and 14C each illustrate a cross-sectional view of the GAA FET device **300** at the stage of fabrication that corresponds to operation **218**. The cross-sectional view of FIG. 14B is cut in a direction along the lengthwise direction of one of the channel structures that remains, e.g., **401A**; and the cross-sectional view of FIG. 14C is cut in a direction along the lengthwise direction of one of the channel structures that has been partially removed, e.g., **401B**.

[0080] The gate structures **1400A** and **1400C** may replace the dummy gate portion **500A** (and remaining sacrificial layers of the channel structure **401A** overlaid by the dummy gate portion **500A**) and the dummy gate portion **500C** (and remaining sacrificial layers of the channel structure **401C** overlaid by the dummy gate portion **500C**), respectively, in accordance with various embodiments. For example, upon forming the isolation structure **1000** and patterning the inner spacers **802I** along the sidewalls of the isolation structure **1000**, the dummy gate portion **500A** is removed. Concurrently or subsequently, the remaining portion of each of the sacrificial layers of the channel structure **401A** is removed, thereby forming a gate trench. Such a trench is filled with the active gate structure **1400A**. The active gate structure **1400A** can wrap around each of the channel layers **420** of the channel structure **401A**, with a pair of the source/drain structures **902** disposed on opposite sides of the active gate structure **1400A** and coupled to the channel layers **420**. Following the similar processes, the active gate structure **1400C** can wrap around each of the channel layers **420** of the channel structure **401C**, with a pair of the source/drain structures **902** disposed on opposite sides of the active gate structure **1400C** and coupled to the channel layers **420**.

[0081] Upon forming the active gate structures, a number of transistors (e.g., GAA FETs) can be formed. For example, a first transistor can be defined by the active gate structure **1400A**, the channel layers **420** of the channel structure **401A**, and a pair of the source/drain structures **902** disposed on the sides of the active gate structure **1400A**; and a second transistor can be defined by the active gate structure **1400C**, the channel layers **420** of the channel structure **401C**, and a pair of the source/drain structures **902** disposed on the sides of the active gate structure **1400C**. With the isolation structure **1000** formed, the first and second transistors (specifically, the active gate structures **1400A** and **1400C**) can be electrically isolated from each other in a relatively small pitch.

[0082] The active gate structures **1400A** and **1400C** can each include a gate dielectric and a gate metal, in some embodiments. The gate dielectric can wrap around each of the corresponding channel layers **420**, e.g., the top and bottom surfaces and sidewalls). The gate dielectric may be formed of different high-k dielectric materials or a similar high-k dielectric material. Example high-k dielectric materials include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The gate dielectric may include a stack of multiple high-k dielectric materials. The gate dielectric can be deposited using any suitable method, including, for example, molecular beam deposition (MBD), atomic layer deposition (ALD), PECVD, and the like. In some embodiments, the gate dielectric may optionally include a substantially thin oxide (e.g., SiO₂) layer, which may be a native oxide layer formed on the surface of each of the channel layers **420**.

[0083] The gate metal may include a stack of multiple metal materials. For example, the gate metal may be a p-type work function layer, an n-type work function layer, multi-layers thereof, or combinations thereof. The work function layer may also be referred to as a work function metal. Example p-type work function metals that may include TiN, TaN, Ru, Mo, Al, WN, ZrSi₂, MoSi₂, TaSi₂, NiSi₂, WN, other suitable p-type work function materials, or combinations thereof. Example n-type work function metals that may include Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. A work function value is associated with the material composition of the work function layer, and thus, the material of the work function layer is chosen to tune its work function value so that a target threshold voltage V_{th} is achieved in the device that is to be formed. The work function layer(s) may be deposited by CVD, physical vapor deposition (PVD), ALD, and/or other suitable process.

[0084] In some other embodiments, the above-discussed isolation structure **1000** can be formed along a channel structure to isolate different portions of the channel structure. FIG. **15** illustrates a top view of the semiconductor device **300** in which an isolation structure **1500** is formed along a channel structure **1501**, in accordance with various embodiments. FIG. **16** further illustrates a corresponding cross-sectional view of the semiconductor device **300** cut along line A-A indicated in FIG. **15** (a lengthwise direction of the channel structure **1501**) Referring to FIG. **15**, the channel structure **1501** extends along the X-direction, with a number of active gate structures **1502A**, **1502B**, and **1502C** straddling respective portions of the channel structure **1501**. The non-straddled portions of the channel structure **1501** can form a number of source/drain structures **1520A**, **1520B**, **1520C**, **1520D**, and **1520E**. The isolation structure **1500**, channel structure **1501**, active gate structures **1502A-C**, and source/drain structures **1520A-E** are substantially similar to the above-discussed isolation structure **1000**, channel structure **401**, active gate structures **1400A-C**, and source/drain structures **902**, respectively. Accordingly, each of the features shown in FIG. **15** will be briefly described as follows.

[0085] For example, a first transistor can be defined by the active gate structure **1502A**, the portion of the channel structure **1501** overlaid by the active gate structure **1502A**, and the source/drain structures **1520A** and **1520B**; a second transistor can be defined by the active gate structure **1502B**, the portion of the channel structure **1501** overlaid by the active gate structure **1502B**, and the source/drain structures **1520B** and **1520C**; and a third transistor can be defined by the active gate structure **1502C**, the portion of the channel structure **1501** overlaid by the active gate structure **1502C**, and the source/drain structures **1520D** and **1520E**. The isolation structure **1500** can electrically isolate the second and third transistors (specifically, the source/drain structures **1520C** and **1520D**).

[0086] As further illustrated in the cross-sectional view of FIG. **16**, the active gate structure **1502A** has its lower portion wrapping around the corresponding channel layers **420**. Such a lower portion of the active gate structure **1502A** is separated (or otherwise isolated) from the source/drain structures **1520A** and **1520B** with a number of inner spacers **802C** discretely disposed along sidewalls of this lower portion. Similarly, the active gate structure **1502B** has its lower portion

wrapping around the corresponding channel layers **420**. Such a lower portion of the active gate structure **1502B** is separated (or otherwise isolated) from the source/drain structures **1520B** and **1520C** with a number of inner spacers **802C** discretely disposed along sidewalls of this lower portion; and the active gate structure **1502C** has its lower portion wrapping around the corresponding channel layers **420**. Such a lower portion of the active gate structure **1502C** is separated (or otherwise isolated) from the source/drain structures **1520D** and **1520E** with a number of inner spacers **802C** discretely disposed along sidewalls of this lower portion.

[0087] The isolation structure **1500**, replacing a portion of the channel structure **1501** interposed between the source/drain structures **1520C** and **1520D**, can then isolate the source/drain structures **1520C** and **1520D**. Specifically, a number of inner spacers **802I** are discretely disposed along sidewalls of the isolation structure **1500**. According to various embodiments of the present disclosure, the inner spacers **802I** can each have its sidewall, facing the isolation structure **1500**, that presents at least one of various profiles as discussed above

[0088] In one aspect of the present disclosure, a semiconductor device is disclosed. The semiconductor device includes a plurality of first channel layers vertically spaced from one another and a plurality of second channel layers vertically spaced from one another. Each of the plurality of first and second channel layers extend along a first lateral direction. The semiconductor device includes an isolation structure disposed between the plurality of first channel layers and the plurality of second channel layers along a second lateral direction perpendicular to the first lateral direction. The semiconductor device includes a plurality of inner spacers discretely disposed along a first sidewall of the isolation structure that faces toward the first lateral direction, or discretely disposed along a second sidewall of the isolation structure that faces away from the first lateral direction wherein an interface between each of the plurality of inner spacers and the first or second sidewall has a vertical profile.

[0089] In another aspect of the present disclosure, a semiconductor device includes a first metal gate structure straddling a first channel structure and a second metal gate structure straddling a second channel structure. The semiconductor device includes an isolation structure disposed between the first and second metal gate structures along a lateral direction perpendicular to a lengthwise direction of the first and second metal gate structures. The semiconductor device includes a pair of first epitaxial structures disposed on opposite sides of the isolation structure, respectively, along the lengthwise direction. The semiconductor device includes a plurality of inner spacers, a first subset of the inner spacers disposed between one of the pair of first epitaxial structures, and a second subset of the inner spacers disposed between the other of the pair of first epitaxial structures, wherein each of the inner spacers has a sidewall interfacing the isolation structure, with a majority portion of the sidewall presenting a vertical profile.

[0090] In yet another aspect of the present disclosure, a method for making a semiconductor device is disclosed. The method includes forming a first fin structure, a second fin structure, and a third fin structure over a substrate, wherein the first to third fin structures are disposed in parallel with one another, with the second fin structure disposed between the first and third fin structures. The method includes forming a dummy gate structure straddling the first to third fin structures. The method includes removing portions of the first fin structure, portions of the second fin structure, and portions of the third fin structure not overlaid by the dummy structure to form a first channel structure, a second channel structure, and a third channel structure, respectively. The method includes forming a plurality of first inner spacers discretely disposed along sidewalls of the first channel structure, a plurality of second inner spacers discretely disposed along sidewalls of the second channel structure, and a plurality of third inner spacers discretely disposed along sidewalls of the third channel structure. The method includes forming a plurality of first inner spacers discretely disposed along sidewalls of the first channel structure, a plurality of second inner spacers discretely disposed along sidewalls of the second channel structure, and a plurality of third inner spacers discretely disposed along sidewalls of the third channel structure.

[0091] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: an isolation structure separating a first gate structure from a second gate structure along a first lateral direction; a first epitaxial structure disposed on a first side of the isolation structure along a second lateral direction perpendicular to the first lateral direction; a second epitaxial structure disposed on a second side of the isolation structure along the second lateral direction; a first inner spacer interposed between the first epitaxial structure and the isolation structure along the second lateral direction; and a second inner spacer interposed between the second epitaxial structure and the isolation structure along the second lateral direction; wherein a first interface between the first inner spacer and the isolation structure and a second interface between the second inner spacer and the isolation structure each have a vertical profile.
2. The semiconductor device of claim 1, further comprising: a plurality of first channel layers vertically spaced from one another, each of the plurality of first channel layers extending along the second lateral direction; and a plurality of second channel layers vertically spaced from one another, each of the plurality of second channel layers extending along the second lateral direction.
3. The semiconductor device of claim 2, wherein the first gate structure wraps around each of the first channel layers, and the second gate structure wraps around each of the second channel layers.
4. The semiconductor device of claim 3, further comprising: a third epitaxial structure disposed on a first side of the first channel layers along the second lateral direction; a fourth epitaxial structure disposed on a second side of the first channel layers along the second lateral direction; a third inner spacer interposed between the third epitaxial structure and each of the first channel layers along the second lateral direction; and a fourth inner spacer interposed between the fourth epitaxial structure and each of the first channel layers along the second lateral direction.
5. The semiconductor device of claim 4, wherein a third interface between the third inner spacer and the corresponding first channel layer and a fourth interface between the fourth inner spacer and the corresponding first channel layer each have a vertical profile.
6. The semiconductor device of claim 3, further comprising: a fifth epitaxial structure disposed on a first side of the second channel layers along the second lateral direction; a sixth epitaxial structure disposed on a second side of the second channel layers along the second lateral direction; a fifth inner spacer interposed between the fifth epitaxial structure and each of the second channel layers along the second lateral direction; and a sixth inner spacer interposed between the sixth epitaxial structure and each of the second channel layers along the second lateral direction.
7. The semiconductor device of claim 6, wherein a fifth interface between the fifth inner spacer and the corresponding second channel layer and a sixth interface between the sixth inner spacer and the corresponding second channel layer each have a vertical profile.
8. The semiconductor device of claim 1, wherein the isolation structure electrically isolates the first gate structure and the second gate structure.
9. The semiconductor device of claim 1, wherein the first inner spacer has a first seam, and the second inner spacer has a second seam.
10. The semiconductor device of claim 9, wherein a first distance laterally measured from a tip of the first seam to the first interface is substantially less than a second distance vertically measured

from the first seam to a top or bottom surface of the first inner spacer.

11. The semiconductor device of claim 9, wherein a first distance laterally measured from a tip of the second seam to the second interface is substantially less than a second distance vertically measured from the second seam to a top or bottom surface of the second inner spacer.

12. A semiconductor device, comprising: an isolation structure separating a first gate structure from a second gate structure along a first lateral direction; a first epitaxial structure disposed on a first side of the isolation structure along a second lateral direction perpendicular to the first lateral direction; a second epitaxial structure disposed on a second side of the isolation structure along the second lateral direction; a first inner spacer interposed between the first epitaxial structure and the isolation structure along the second lateral direction; and a second inner spacer interposed between the second epitaxial structure and the isolation structure along the second lateral direction; wherein a first interface between the first inner spacer and the isolation structure and a second interface between the second inner spacer and the isolation structure each have a vertical profile, wherein the first and second epitaxial structures and the isolation structure each have a curvature-based bottom surface.

13. The semiconductor device of claim 12, further comprising: a plurality of first channel layers vertically spaced from one another, each of the plurality of first channel layers extending along the second lateral direction; and a plurality of second channel layers vertically spaced from one another, each of the plurality of second channel layers extending along the second lateral direction.

14. The semiconductor device of claim 13, wherein the first gate structure wraps around each of the first channel layers, and the second gate structure wraps around each of the second channel layers.

15. The semiconductor device of claim 13, further comprising: a third epitaxial structure disposed on a first side of the first channel layers along the second lateral direction; a fourth epitaxial structure disposed on a second side of the first channel layers along the second lateral direction; a third inner spacer interposed between the third epitaxial structure and each of the first channel layers along the second lateral direction; and a fourth inner spacer interposed between the fourth epitaxial structure and each of the first channel layers along the second lateral direction.

16. The semiconductor device of claim 15, wherein a third interface between the third inner spacer and the corresponding first channel layer and a fourth interface between the fourth inner spacer and the corresponding first channel layer each have a vertical profile.

17. The semiconductor device of claim 13, further comprising: a fifth epitaxial structure disposed on a first side of the second channel layers along the second lateral direction; a sixth epitaxial structure disposed on a second side of the second channel layers along the second lateral direction; a fifth inner spacer interposed between the fifth epitaxial structure and each of the second channel layers along the second lateral direction; and a sixth inner spacer interposed between the sixth epitaxial structure and each of the second channel layers along the second lateral direction.

18. The semiconductor device of claim 17, wherein a fifth interface between the fifth inner spacer and the corresponding second channel layer and a sixth interface between the sixth inner spacer and the corresponding second channel layer each have a vertical profile.

19. A semiconductor device, comprising: a first metal gate structure extending along a first lateral direction and wrapping around each of a plurality of first channel layers extending along a second lateral direction; a second metal gate structure extending along the first lateral direction and wrapping around each of a plurality of second channel layers extending along the second lateral direction; an isolation structure separating the first metal gate structure from the second metal gate structure; a first epitaxial structure disposed on a first side of the isolation structure along the second lateral direction; a second epitaxial structure disposed on a second side of the isolation structure along the second lateral direction; a first inner spacer interposed between the first epitaxial structure and the isolation structure along the second lateral direction; and a second inner spacer interposed between the second epitaxial structure and the isolation structure along the second lateral direction; wherein a first interface between the first inner spacer and the isolation

structure and a second interface between the second inner spacer and the isolation structure each have a vertical profile.

20. The semiconductor device of claim 19, wherein at least one of the first or second inner spacer has a scam.
